MEC E 686 Assessment and Analysis of Biomechanical Motion

DEPARTMENT OF MECHANICAL ENGINEERING
UNIVERSITY OF ALBERTA

Winter 2026
Instructor: Dr. Hossein Rouhani

Assignment 3 (*100 marks*)

Important assignment information: This assignment is due on **Thursday, April 9, 2026.** Please submit a single PDF file of your assignment solution via the Canvas course webpage before 11:00 am on this date. In completing the assignment, **students must provide complete solutions**: the process used in answering the assignment questions should be described, and any MATLAB code developed as part of the solution process should be referenced and included in an appendix. **Using the information contained in the files included with the previous assignments as well as the Biomechanics Assignments Overview document, answer the following questions:**

1. (***15 marks***) For each trial and muscle, process the electromyogram (EMG) data included with Assignment 2 by: (1) filtering it using a fourth-order, zero phase-shift, high-pass Butterworth filter with a cut-off frequency of 10 Hz; (2) filtering it using a fourth-order, zero phase-shift, low-pass Butterworth filter with a cut-off frequency of 500 Hz; (3) demeaning it; and (4) rectifying it. The filtering should be done using the MATLAB commands described in Question 9 of Assignment 1. **Note that you should <u>not</u> provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be referenced and included in an appendix.**
2. (***5 marks***) For each trial and muscle, calculate the standard deviation (SD) of the processed muscle activity in the first second of the trial (i.e., the quiet standing period of the trial). **Note, again, that you should <u>not</u> provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be referenced and included in an appendix.**
3. (***5 marks***) For each trial, downsample the processed EMG data from 2,000 to 100 Hz (i.e., so that its sampling frequency matches that of the motion capture data) by extracting the EMG samples with Frames and Sub-Frames that match those of samples in the motion capture data included with Assignment 1 (i.e., you should extract rows 1, 21, 41, etc. in the EMG data for each trial). **Note that you should <u>not</u> provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be referenced and included in an appendix.**
4. (***5 marks***) Using the results of Question 3 as well as of Question 4 in Assignment 1, **for each trial where the right foot landed on the force plate**, extract the processed EMG data for the force plate gait cycle. For example, for the first trial of Participant 1, you should extract the

EMG data from samples 332 to 459 and discard the rest of the EMG data for this trial. **Note, again, that you should not provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be referenced and included in an appendix.**

5. (***5 marks***) **Note: For this question and the following one, analyze only the data extracted in Question 4.** Using the results of Question 4, use linear interpolation to time normalize the EMG time series for each trial to 101 samples. This should be done using the MATLAB command described in Question 6 of Assignment 1. **Note, again, that you should not provide the results of these calculations; instead, the process used to obtain them should be described, and associated MATLAB code should be referenced and included in an appendix.**

6. (***50 marks***) Using the results of Questions 2 and 5 as well as of Question 7 in Assignment 1, for each participant, create a 3-by-1 figure that presents: (1) the ensemble average flexion/extension (F/E) angle of the right knee (top subplot); (2) the ensemble average F/E angle of the right ankle (middle subplot); and (3) for each muscle and trial, a visual representation of the periods where the muscle activity exceeded three times its standard deviation during quiet standing (i.e., the periods where the muscle was 'active') (bottom subplot). The ensemble average angles should be presented, in units of degrees, as the mean ±1 SD bounds. The figure for Participant 1 is provided in Figure 1 to illustrate how the figures should look. **Provide the created figures (three figures total). The MATLAB code used to create the figures should be referenced and included in an appendix.**

7. (***15 marks***) Based on a visual comparison of the joint angles to the periods where each muscle was active (for Participant 1 only), provide an explanation for the role of each muscle during gait. In your explanation, please also comment on aspects of muscle activation variability across trials as well as on aspects of single muscle contraction versus co-contraction of agonist and antagonist muscles (for a given joint).

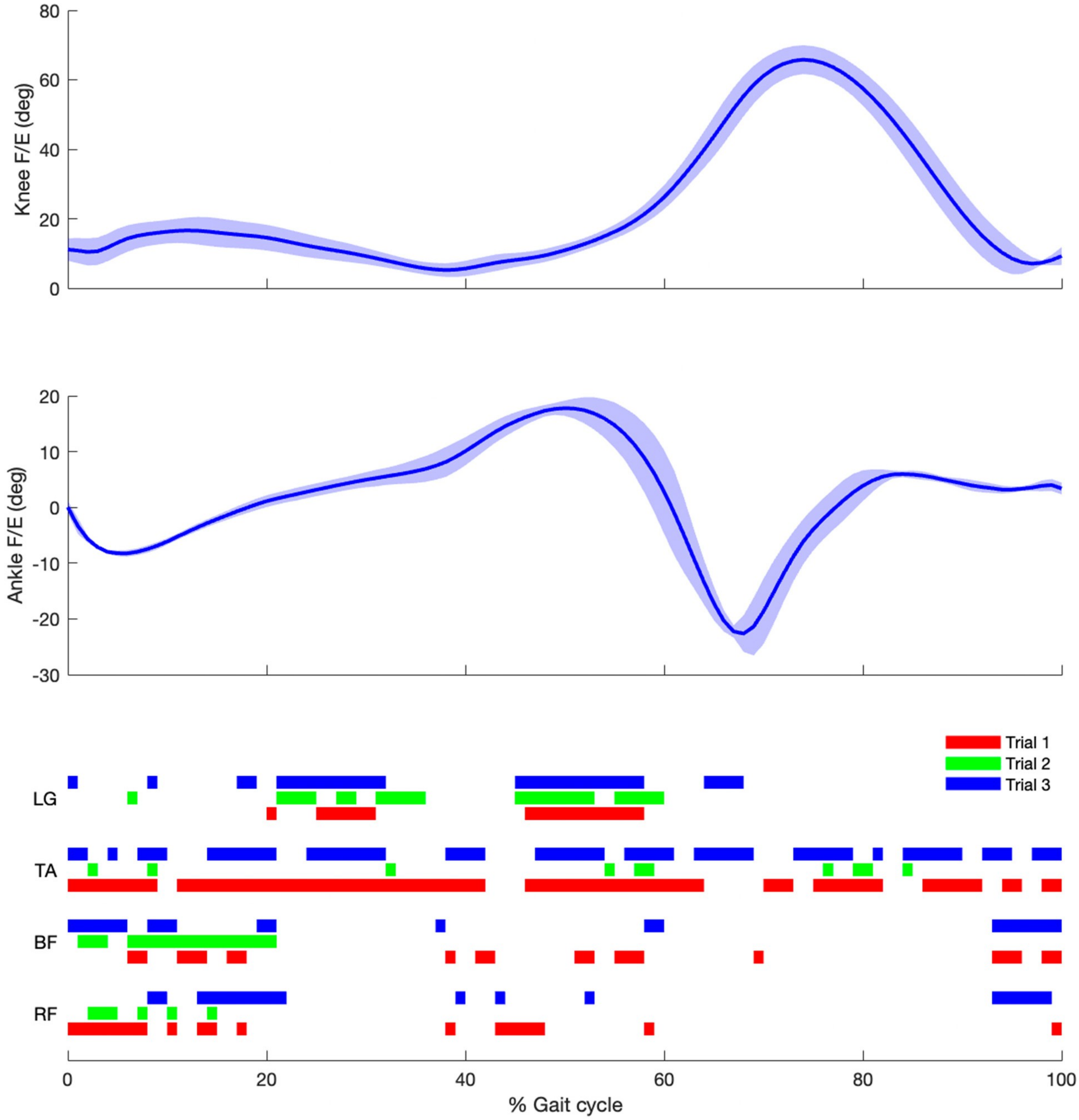

Figure 1: Ensemble average (mean ±1 standard deviation bound) flexion/extension (F/E) angle of the right knee (top subplot) and ankle (middle subplot) as well as a visual representation of the periods where the activity of the lateral gastrocnemius (LG), tibialis anterior (TA), biceps femoris (BF), and rectus femoris (RF) exceeded three times their standard deviations during quiet standing (bottom subplot) for Participant 1.